King Saud University
College of Computer and Information Sciences
Department of Information Technology

Semester-2 1447H (Spring 2026)

Graduation Project Proposal

Gaming Shield

Prepared by

Malak Talal Basloom, 444204601
Rathath Bin Rakan, 444200450
Hatoon Alghamdi , 444200480
Nora Alkhudair, 444201090

Supervised by
Dr. Rana Almurshed

Second Semester 1447 H Spring 2026

Table of Contents

1 Introduction 3
2 The Problem 4
3 The Solution 5
4 Product Vision 5
5 Product Roadmap 6
6 Objectives 6
7 Scope 8
8 Hardware/Software Tools and Cost 10
9 Scrum Team 11
9.1 Skill Set Requirements 11
9.2 Roles and Responsibilities 12
10 References 13

1 Introduction

In recent years, online video games have evolved from mere entertainment to interactive digital environments in which millions of players worldwide communicate, collaborate, and compete directly across various platforms. These games are no longer solely individual activities but have become virtual social platforms reflecting real human interactions among people of different cultures and ages. While this interaction provides opportunities for learning and social bonding, it also opens the door to negative digital behaviors that can affect users' safety and psychological well-being, particularly among children and adolescents **[1].**

Video games enjoy widespread popularity both locally and globally, with children and young people making up a significant portion of users. For many families, age-rated games are considered a safe recreational option for their children. However, the reality within these games may differ from official ratings, as studies indicate that multiplayer game environments can involve behaviors such as cyberbullying, harassment, hate speech, and other harmful interactions. Reports have shown that a noticeable proportion of players have experienced harassment or negative incidents while playing online **[2].** This highlights a clear gap between theoretical age ratings and the actual experience of players within interactive environments.

The problem is not limited to the accuracy of age ratings; it extends to a more serious issue, where many games containing harmful interactive environments remain available for long periods without proper risk monitoring. Current monitoring mechanisms rely heavily on scattered individual reports or manual reviews, which delays the detection of dangerous behavioral patterns. Consequently, children may continue using games with offensive or harmful content without the knowledge of parents or relevant authorities, and the absence of a unified platform that collects, analyzes, and clearly presents these reports creates a gap between users and regulators, making the response slower and less effective.

Technology provides an effective opportunity to bridge this gap. By leveraging artificial intelligence and natural language processing, it is possible to analyze large volumes of user reviews and comments on different gaming platforms, such as app stores and distribution platforms, where users express their experiences, including exposure to bullying, toxic behaviors, or inappropriate content. This type of data provides an indirect yet wide-ranging understanding of the interactive environment of each game. At the local level, the Kingdom of Saudi Arabia, within the framework of Vision 2030, seeks to enhance the role of video games and digital technologies in developing the digital economy and fostering local skills, while supporting innovation and creating a safe and attractive environment for Saudi youth in this sector **[3].**

This project proposes the development of a smart platform that measures and displays the risk level of each game based on the analysis of public reviews and comments, in addition to user reports within the platform itself. The system calculates a dynamic risk score for each game and identifies the detected risk categories, such as bullying, hate speech, inappropriate content. The platform also serves as a link between users and governmental authorities by aggregating player reports and clearly presenting high-risk games, helping relevant authorities respond more quickly

and accurately. This transforms scattered individual reports into organized data that supports evidence-based decision-making.

The importance of this solution lies in its service to multiple stakeholders: it helps parents choose safer games for their children, enables regulatory authorities to monitor digital risks early, and contributes to improving the digital gaming environment, making it safer for young users. At the global level, this approach represents a shift from reactive monitoring, which relies on a large number of complaints before action, to a proactive method based on continuous data analysis. By combining community reports with intelligent content analysis, the platform helps build a safer and more transparent digital environment for future generations.

2 The Problem

In the rapid expansion of digital gaming, these environments have evolved beyond visual gameplay into highly interactive social spaces. This evolution has exposed children to significant risks through player-to-player communication and unregulated user-generated content. Current monitoring mechanisms are largely reactive, depending on manual reports and the accumulation of complaints before any action is taken. This delay allows inappropriate language, violent discussions, and harmful ideological influences to persist, often resulting in psychological harm before authorities or parents can intervene.

The scale of this issue is reflected in global and local statistics. According to a study by the Anti-Defamation League (ADL), about 62% of young gamers (ages 13-17) reported experiencing harassment while playing online multiplayer games **[4]**. Locally, in Saudi Arabia, approximately 47% of children reported experiencing or witnessing online bullying within video games **[5].** These figures highlight that nearly half of the child population in gaming environments is directly exposed to harmful interactions that current filters fail to catch in real-time.

This project specifically tackles the Reactive Monitoring Gap : a monitoring limitation in which oversight relies on delayed reactions, such as user complaints, instead of proactive complaints.

Instead of waiting for a complaint to be filed after the harm has occurred, our project focuses on using Artificial Intelligence (AI) and Natural Language Processing (NLP) to proactively analyze player interactions and comments.

The platform specifically targets the protection of children and adolescents, with a primary focus on 7 years and above age group, who are statistically the most active and exposed to social risks in multiplayer environments**.** We aim to transform child protection from a manual, slow process into an automated, dynamic system that detects emerging risks (like extremist content or cyberbullying) immediately, providing early warnings to both parents and government authorities.

3 The Solution

To address the limitations of reactive monitoring and the lack of real-time visibility in digital gaming environments, this project proposes an **AI-driven intelligent web platform** that analyzes game-related data collected from multiple online sources. Rather than relying on manual user complaints or direct in-game access, the system aggregates information from publicly available platforms such as game reviews, player comments, online discussions, and curated datasets.

Using **Artificial Intelligence (AI)** and **Natural Language Processing (NLP)**, the platform processes large volumes of text-based data to detect patterns related to cyberbullying, hate speech, extremist ideologies, violent discussions, and other harmful content. By applying trained classification models, the system assigns **risk levels** to each game .

The platform provides **clear dashboards, alerts, and risk summaries** tailored to different users:

- **Government authorities** access real-time overviews, detailed risk breakdowns, and prioritized case management tools.
- **Individual users** can follow games, view risk for each game, compare risks, and receive early warnings about emerging threats.

By transforming scattered online information into structured, real-time intelligence, the platform closes the "reactive monitoring gap" and enables earlier detection of harmful trends before they escalate into serious psychological or social harm.

4 Product Vision

Product Vision:

For an individual users concerned about safety in digital games.

Who needs a reliable way to detect and understand risks such as cyberbullying, violent

content, and harmful ideologies in online gaming environments.

The Gaming Shield is a web platform .

That analyzes publicly available game-related data to provide risk assessment, and age-appropriate recommendations .

Unlike traditional platforms such as Steam Reviews.

Our product delivers AI-driven risk intelligence across gaming ecosystems by analyzing data from multiple internet sources .

5 Product Roadmap

6 Objectives

This section outlines the objectives of the graduation project, categorized into product, project, and learning objectives.

A- Product Objectives (Customer-Focused Value)

1. **Provide AI-Based Game Risk Monitoring**
   Develop a platform that analyzes digital games and player comments using AI to detect risks such as harassment, violent language, and inappropriate communication in.
2. **Improve Parental Awareness and Decision-Making**
   Enable parents and individual users to easily track games, view clear risk levels, age suitability, and receive alerts, helping them make safer gaming choices for children.
3. **Deliver Risk Alerts**
   Notify users when a tracked game's risk level changes, ensuring early awareness before harmful content escalates.
4. **Support Government and Regulatory Monitoring**
   Provide government employees with dashboards to monitor high-risk games, analyze trends, and manage reports to support faster and more informed regulatory action.
5. **Replace Static Ratings with Dynamic Risk Classification**
   Complement traditional age ratings (e.g., PEGI) with a dynamic, AI-driven risk score that updates based on real-time player-generated content.
6. **Enable Transparent Game Comparison**
   Allow users to compare games based on risk level, popularity, and category to identify safer alternatives.
7. **Promote Digital Safety Awareness**
   Increase awareness of hidden online gaming risks by presenting analytics, statistics, and risk explanations in a clear and accessible way.

B- Project Objectives (Solution-Focused Plan)

1. **Analyze User Needs and System Requirements**
   Identify requirements for parents, government employees, and administrators based on real-world digital safety challenges.
2. **Design a Multi-Role Platform Architecture**
   Implement role-based access control for parents, government employees, and administrators.
3. **Collect and Prepare Game-Related Data**
   Gather datasets such as game descriptions, reviews, and player comments, then clean and preprocess the data for analysis.
4. **Develop an AI-Based Risk Detection Model**
   Train and test an NLP-based model to classify content into different risk categories (e.g., low, medium, high risk).
5. **Build Interactive Dashboards and Interfaces**
   Design user-friendly dashboards for monitoring risks, tracking games, managing reports, and viewing analytics.

6. **Implement a Alert System**
   Create a notification mechanism that sends alerts when new risks are detected in games.
7. **Test, Validate, and Deploy the System**
   Perform usability testing, AI accuracy evaluation, and system testing to ensure reliability and scalability

C- Learning Objectives (Student-Focused Growth)

1. **Applied AI and NLP for Content Moderation**
   Gain practical experience in building and evaluating AI models for real-world safety applications.
2. **Full-Stack Platform Implementation**
   Strengthen skills in frontend design, backend development, database integration, and authentication systems.
3. **Multi-User Experience Design**
   Learn how to design interfaces tailored to different user roles with varying goals and permissions.
4. **Ethical and Responsible AI Development**
   Understand ethical considerations such as bias, privacy, and transparency when working with AI and sensitive user data.
5. **Integration of External Platforms and Standards**
   Gain experience working with external data sources and industry standards such as age rating systems.
6. **End-to-End Project Execution**
   Improve planning, documentation, testing, and presentation skills required for a complete graduation-level project.

7 Scope

This project aims to design and develop a web-based platform that leverages Artificial Intelligence (AI) to analyze textual content related to online PC games and assess their risk levels. The system will focus on user-generated content such as game reviews and comments to provide a data-driven evaluation of game safety for players aged 7 years and above.

The platform will be accessible through modern web browsers and will utilize widely adopted web and AI technologies, primarily Natural Language Processing (NLP) for text analysis. Direct integration with in-game chats or real-time voice communication is outside the current scope.

The high-level boundaries of the project include:

- **Technology and Platform:** Web-based application accessible via desktop and mobile browsers; no native mobile apps.
- **Language Support:** English textual content only; other languages may be considered in future versions.
- **Data Sources:** Publicly available game reviews and comments; private data or real-time in-game communications are excluded.
- **User Roles:** Parents/individual users, government employees, and system administrators with predefined access permissions.
- **Artificial Intelligence Capabilities:** Classification of textual content into risk levels (Low, Medium, High); no advanced behavioral analysis.

This scope ensures feasibility within the academic timeframe while delivering a functional solution that helps users identify and avoid risky PC games.

8 Hardware/Software Tools and Cost

Hardware Tools	
Name and Description	Cost *mention date it will be available*
Personal Computers / Laptops	Already available – No additional cost (Available from project start)
Software Tools	
Name and Description	Cost
Web Development Tools (HTML, CSS, JavaScript)	Free
Backend Framework (PHP / Node.js)	Free
Backend Framework (PHP / Node.js)	Free
GPU / Compute Services – Used for AI model training and large-scale data processing.	SAR 40 – 115 per month
Paid APIs – Used to retrieve game data, reviews, and comments at scale.	SAR 75 – 375 per month
Domain Name – Used to publish the SafePlay platform online.	SAR 40 – 60 per year
Paid Dataset (Optional) – Used to enhance AI model training and improve classification accuracy.	SAR 200 – 750 (one-time cost)
Generative Ai subscription (chatGPT , gemini , others...)	SAR 75 – 200 per month

Other Relevant Costs

- **Cloud and Compute Costs:**
  Monthly cloud hosting and GPU usage represent ongoing operational expenses.
- **API Usage Costs:**
  Costs may increase depending on the volume of API requests and data retrieval.
- **Dataset Acquisition Costs:**
  A one-time cost may be incurred if premium datasets are used to improve AI accuracy.
- **Generative Ai subscription**

  Costs may increase depending on the AI using to improve the model and the other project requirements.

Note: The total project cost may exceed **SAR 1000**, mainly due to cloud hosting, GPU usage, and paid APIs, while hardware costs remain minimal, as stated in the GP1 document.

9 Scrum Team

9.1 Skill Set Requirements

Technical Skill Required	What is the current level of the team (*beginner-intermediate- advanced*) for each skill? How will the gap be bridged? *(if necessary*) Learning plan
Full-Stack Web Development	Intermediate
Database Management	Intermediate
API Integration	Beginner, Studying the official documentation for each API to ensure smooth data collection of game reviews and comments.
AI & Natural Language Processing	Beginner, Self-paced learning through online courses and practicing with the Kaggle dataset.
Data Analysis & Visualization	Intermediate, Learning how to use libraries to transform AI results into clear percentage-based breakdown charts.
Cloud Hosting & Deployment	Beginner, Self-learning using official cloud documentation and practical tutorials to set up the server, database, and ensure 24/7 online access.

9.2 Roles and Responsibilities

Scrum Team	
Product Owner: (list name of the project supervisor)	Dr. Rana Almurshed
Developers: (list student names)	Hatoon Alghamdi Nora Alkhudair Rathath Bin Rakan Malak Talal Basloom
Scrum Master (SM): (list name of designated SM for the project)	Dr. Rana Almurshed
Stakeholders: (list names of project stakeholders, include customer names if any)	Graduation Project Committee

10 References

[1] Hu, Y., Sophie, E., Clancy, E. M., & Klettke, B. (2025). *Potential negative effects of multiplayer online games: A systematic review of cyberbullying* [Research article]. *Computers in Human Behavior Reports*. Retrieved from https://doi.org/10.1016/j.chbr.2025.100675

[2] Anti-Defamation League. (2023). *Hate Is No Game: Hate and Harassment in Online Games* [Report]. Retrieved from https://www.adl.org/resources/report/hate-no-game-hate-and-harassment-online-games-2023

[3] Saudi Gazette. (2025). Crown Prince launches National Gaming and Esports Strategy. Retrieved from https://saudigazette.com.sa/article/625009/SAUDI-ARABIA/Crown-Prince-launches-National-Gaming-and-Esports-Strategy

[4] *Almost two-thirds of young gamers experience harassment while gaming, says study*. (2021, October 27). *NME*. Retrieved from https://www.nme.com/news/almost-two-thirds-of-young-gamers-experience-harassment-while-gaming-says-study-3080394

[5] Arab News. (2021). *Saudi Arabia: 47% of children in Saudi Arabia have come across cyberbullying***.** Jeddah, SA: Arab News. https://www.arabnews.pk/node/1827856/saudi-arabia

Note: AI was used to assist in drafting and refining parts of this document to ensure clarity and quality.